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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution Affiliated to VTU)

III Semester B. E. Regular / Supplementary Examinations Jan / Feb-2025

Common to All Programs

ENVIRONMENT AND SUSTAINABILITY

Time: 03 Hours *Maximum Marks: 100*

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

PART-A

			M	BT	CO
1	1.1	Highlight any two strategies of conserving the ecosystem.	02	1	1
	1.2	Enumerate the 10% rule of energy flow.	02	1	1
	1.3	How Renewable Energy are sustainable?	02	2	1
	1.4	Highlight the principle of Green Engineering.	02	1	2
	1.5	Define Environmental economics.	02	1	3
	1.6	Recall the Goal seven of Millennium Development Goals.	02	1	2
	1.7	Enlist any four activities to control Climate Change.	02	1	3
	1.8	Highlight the concept of carbon offset.	02	1	3
	1.9	What is life cycle assessment of a material?	02	1	3
	1.10	State the 4 pillars of *CSR*.	02	1	4

PART-B

2	a	Describe pollutants. Enumerate the different types of pollution.	08	2	2
	b	Define *EMS*. How *EMS* is structured? Explain.	08	2	3
3	a	Justify how renewable energy benefit Environment and save money.	08	5	1
	b	With a neat sketch of carbon cycle enumerate the steps involved in the same.	08	2	1
		OR			
4	a	Distinguish between conventional and non-conventional source of energy.	08	2	2
	b	How Transit Oriented Development (*TOD*) promotes sustainable development. Justify.	08	5	3
5	a	Discuss the scope and goals of Environmental Auditing.	08	2	3
	b	Illustrate the economic and environmental benefits of a circular economy.	08	3	3
		OR			
6	a	State and explain the key features of transition from a linear to circular economy.	08	2	3
	b	Enumerate the concept of Industrial ecology.	08	3	3

7	a	With reference to Global Climate change highlight water scarcity and causes of water scarcity.	08	2	2
	b	Interpret how carbon credits is valuable tool in combating climate change.	08	5	3
		OR			
8	a	Define *EIA*? Enumerate the benefits of conducting *EIA*.	08	2	2
	b	Discuss the evolution of Life cycle assessment.	08	2	3
9	a	Summarize the key aspects of *CSR*.	08	2	4
	b	Discuss in brief the concept of corporate Philanthropy.	08	2	4
		OR			
10	a	Enumerate the Triple bottom line concept for sustainable business.	08	3	4
	b	Illustrate the broader relationship between *CSR* and Corporate Governance.	08	3	4

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU
III Semester B. E. Examinations MARCH-2024

ENVIRONMENT AND SUSTAINABILITY (BASKET COURSE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8 & 9 and 10.

PART-A

1	1.1	Define the terms Eco-system and Bio-diversity.	02
	1.2	List various Energy cycles in nature.	02
	1.3	Illustrate what is carbon sequestration.	02
	1.4	Mention the significance to stop man wildlife conflicts.	02
	1.5	Describe the importance of GDP.	02
	1.6	Highlight any four threats to Bio-diversity.	02
	1.7	Mention any two major causes for climate changes.	02
	1.8	Suggest the significance of EIA as a tool for sustainable practices.	02
	1.9	What is performance measurement in business.	02
	1.10	Enumerate the prominence of CSR.	02

PART-B

2	a	Classify different types of ecosystems. Explain in brief any two ecosystems with their structure and composition.	08
	b	Describe the idealization made in Engineering Mechanics and also the characteristics of a force.	08
3	a	Explain carbon cycle with sketch.	08
	b	Enumerate the concept of sustainable urbanization and the need for the same.	08
		OR	
4	a	Explain ocean thermal energy conversion with its advantages and limitations.	08
	b	Explain nitrogen cycle with sketch.	08
5	a	What is environmental audit? Explain the steps in environmental audit.	08
	b	Enumerate the economic and social aspects of sustainability.	08
		OR	
6	a	Explain the linear and cyclic resource management systems.	08

b	What are the advantages and limitations of industrial ecology?	08

7	a	Define the following i) Climate change ii) Carbon credit iii) Carbon footprint iv) Environmental impact assessment.	08
	b	Explain the concept of 5R.	08
		OR	
8	a	Enumerate ISO 14000 series standards.	08
	b	Define green buildings, enumerate the green building materials and write the advantages of green buildings.	08
9	a	Explain the principles of CSR.	08
	b	Explain the concept of charity and corporate philanthropy.	08
		OR	
10	a	Enumerate the environmental aspect of CSR.	08
	b	Explain how CSR has evolved in India.	08

RV College of Engineering

Semester B.E. Examinations – April/May 2024
ENVIRONMENT AND SUSTAINABILITY

Time: 3 Hours
Maximum Marks: 100

Instructions to Candidates:

1. Answer all questions from Part A. Part A questions should be answered in the first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B, Question Number 2 is compulsory. Answer any one full question from 3 & 4, 5 & 6, 7 & 8, 9 & 10.

PART A

1. Define the terms Biotic and Abiotic components.
2. Mention threats to biodiversity.
3. List out the types of land pollution.
4. State the effects of noise pollution.
5. What are fossil fuels and why are they non-renewable?
6. What is biogas and mention its uses.
7. Highlight the significance of GDP in national economic activity.
8. Highlight the distinction between Kyoto Protocol and Paris Agreement.
9. Mention the principles of organic farming.
10. Enumerate the significance of (EIA) in developmental projects.

PART B

2. a) Discuss the importance of any four environmental protection acts promulgated in India.
   b) Briefly discuss the causes, effects, and preventive measures for soil and air pollution.

3. a) Explain the concept of energy management and conservation.
   b) Enumerate the concept of sustainable urbanization and the need for the same.

OR

4. a) Explain the concept of hydrogen energy with its uses, advantages, and disadvantages.
   b) Explain the carbon cycle with sketch.

5. a) Define environmental audit and briefly describe the types of environmental audit.
   b) Enumerate the economic and social aspects of sustainability.

OR

6. a) Explain the key concepts of system thinking along with its advantages.
   b) Discuss the concepts of circular economy and industrial ecology.

7. a) Enumerate the various ways of reducing carbon footprint in our daily life.
   b) State and explain how green building concept evolved as a sustainable habitat.

OR

8. a) Enumerate IS 14000 series of standards.
   b) Explain in brief the necessity of energy efficiency and sustainable transport.

9. a) Explain the principles of CSR.
   b) Explain the concept of charity and corporate philanthropy.

OR

10. a) Enumerate the environmental aspects of CSR.
    b) Explain how the changing emphasis of companies focusing on CSR.

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution Affiliated to VTU)

IV Semester B. E. Examinations Sept./Oct. – 2024

Civil Engineering

ENVIRIONMENT AND SUSTAINABILITY (BASKET COURSE)

Time: 03 Hours *Maximum Marks: 100*

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

PART-A

			M	BT	CO
1	1.1	List the components of environment.	02	1	1
	1.2	Differentiate between primary and secondary succession.	02	4	1
	1.3	Define climate change.	02	1	2
	1.4	List any two renewable and non renewable sources of energy.	02	1	2
	1.5	Define GDP as an economic indicator.	02	1	3
	1.6	Enlist any four Millennium Development Goal (MDG's).	02	1	3
	1.7	List any four ozone depleting substances.	02	1	2
	1.8	Define carbon foot print for a product or process.	02	1	3
	1.9	Write the principles of CSR.	02	2	4
	1.10	What are the two examples of business ethics?	02	2	4

PART-B

2	a	Classify different types of ecosystems. Explain in brief any two ecosystems with their structure and composition.	08	2	1
	b	Briefly describe the various Environmental protection acts promulgated in India.	08	2	2
3	a	Demonstrate on "Hydrogen energy is a clean energy", write the advantages and disadvantages of Hydrogen energy as an energy source.	08	2	2
	b	Enumerate the process of energy management and energy conservation with its objectives.	08	2	2
		OR			
4	a	Explain : i. Biomass energy ii. Ocean thermal energy conversion	08	2	2
	b	Define Energy cycle and explain with a neat sketch carbon cycle.	08	2	2
5	a	Discuss Environmental Economics and Environmental audit as a measure to sustainability.	08	2	3
	b	Discuss linear resource management systems as bane in sustainable Growth of industries.	08	2	3
		OR			

6	a	Illustrate how sustainable development is concept of three aspects of society.	08	2	2
	b	Define industrial ecology and explain the strategies to implement the same.	08	2	2
7	a	Enumerate all the 17 SDG's stipulated by United nations and brief the significance of any one.	08	1	3
	b	Explain Zero waste concept as a philosophical approach.	08	2	3
		OR			
8	a	Describe Circular Economy and distinguish linear vs circular economy.	08	2	3
	b	Explain ISO 14000 and carbon Sequestration.	08	2	3
9	a	Recall and explain the history of evolution of CSR.	08	2	4
	b	Illustrate the concept of charity and corporate philanthropy.	08	2	4
		OR			
10	a	Discuss the concept of sustainability and stakeholder Management.	08	2	4
	b	Enumerate the key features of Global Reporting Index and Dow Jones sustainability index.	08	1	4